

CarbonPass

A conversational, local-first AI that turns Taiwan's open data into verified market access for 2,600 exporting SMEs.

Closing the carbon-data divide, starting in the Kingdom of Screws.

2026 Presidential Hackathon — International Track

Theme: "Digital Inclusion in the AI Era"

Corresponding SDGs: 8 · 9 · 12 · 13

Supervisor: Office of the President, ROC (Taiwan)

Implementer: Ministry of Digital Affairs

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Every quantitative claim in this proposal is cited to a primary or dated, credible source. Figures that are estimates are labeled as estimates; pending legislation is presented as pending. All content is submitted in English.

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Executive Summary

On 1 January 2026 the European Union began charging importers for the carbon embedded in imported steel goods. In the preceding reporting period Taiwan shipped **3.74 million tonnes** of covered goods to the EU — mostly screws and fasteners produced by **about 2,600 small, family-run factories** concentrated in the Gangshan district of Kaohsiung, the “Kingdom of Screws”¹. From this year each of those factories must either deliver verified, product-level carbon data to its European buyer, or let that buyer absorb punitive “default values” — marked up 10% now and 30% by 2028⁸ — until the order moves to a supplier who can supply the data.

The information needed to avoid that penalty exists, but it is buried in a 2,400-page EU implementing regulation⁸ that a senior factory owner, whose back office is a family member with a spreadsheet, cannot practically read — and no tool in Taiwan’s current public or commercial ecosystem produces the required artifact for him^{1,18,20}. This is the digital divide of the AI era in its most economically acute form: not the absence of connectivity, but the inability to produce the data that now decides who may participate in global markets.

CarbonPass closes that divide from a photograph. Inside LINE, in Mandarin or Taiwanese, an owner photographs an electricity bill and a set of invoices. A local-first AI — raw data never leaves the factory — returns three outputs: (1) the verifier-ready, per-product carbon data pack the EU buyer requires, in the European Commission’s own template; (2) a decision screen quantifying what the data is worth, “*with your figures the buyer pays $\approx$ EUR 150 per tonne; without them, EUR 450–750 and rising*”^{17,23}; and (3) a production schedule, computed from Taiwan’s open 10-minute grid data²⁶, that shifts flexible loads to cheaper, lower-carbon hours — reducing the electricity bill now and the next data pack’s emissions with it.

The proof of concept is built entirely on Taiwan’s open data (MOENV emission coefficients, Taipower grid telemetry and tariffs) and the EU’s published methodology, and it gives data back: a cleaned hourly grid-carbon-intensity dataset and an anonymized sector benchmark, openly licensed. The beachhead population is precisely defined — 2,600 CBAM-exposed exporters within a $\sim$ 1,800-firm, top-three world export cluster — and the same engine scales as the mechanism expands: the European Parliament votes in September 2026 on extending coverage to $\sim$ 180 further product categories¹⁰, and, beyond Taiwan, every SME-exporting economy faces the same wall.

1 Team Information

Application form field 1. Member names, nationalities, contact emails, phone numbers, gender, and current country of residence are entered in the online application form; the primary and secondary contacts are listed first, as required.

TEAM COMPOSITION AND ELIGIBILITY

The team comprises between 3 and 10 members and includes at least one member holding nationality other than that of the Republic of China (Taiwan), in accordance with the eligibility rules²⁹. Two members are designated as primary and secondary contacts. The team spans four competencies matched to the work: machine learning and document intelligence; energy-systems and optimization engineering; EU/Taiwan carbon-regulation analysis; and fastener-industry domain knowledge. All project content, documentation, and presentation materials are prepared and submitted in English.

ORIGINALITY AND COMMITMENTS

The entry is the original work of the team, created for this competition. It has not been sold or awarded elsewhere, so the 50% modification rule does not apply. Intellectual-property arrangements among members follow the Handbook's terms²⁹. The team commits to full participation in the mentorship period, implementation-result presentations, and follow-up benefit tracking required of advancing teams.

2 Project Information

2.1 Project name and introduction

Project name: CarbonPass — a local-first AI compliance and energy-optimization copilot for Taiwan's export SMEs.

CarbonPass is a conversational artificial-intelligence system that converts the ordinary documents a small factory already holds — electricity bills, material invoices, and production records — into the exact carbon-compliance artifact its European customers now require, and then schedules the factory's energy use to cut both cost and emissions. The user experience is a single sentence: *photograph an electricity bill and invoices inside LINE, answer three questions in Mandarin or Taiwanese, and receive (a) the verifier-ready carbon data pack the EU buyer needs, (b) the money that data saves the buyer, and (c) a production schedule that lowers the power bill and improves next quarter's emissions.*

DETAILED DESCRIPTION OF AI USAGE (APPLICATION FORM FIELD 2.2)

CarbonPass is an AI system, not an arithmetic calculator with an interface. Three distinct AI/ML subsystems carry the product:

(1) Document intelligence with allocation reasoning. A vision-language model parses heterogeneous, mixed-language, photographed documents (Taipower bills, Ministry of Finance electronic invoices, handwritten production logs) into structured activity data. The non-trivial machine-learning problem is *allocation*: splitting one facility-wide electricity figure across dozens of product variants and customs (CN) codes, which the EU's product-level methodology strictly requires but invoices never contain. The system infers allocation as a constrained-inference problem over production volumes, machine power ratings, and operating-hour priors, and attaches a *quantified uncertainty* to every line so a human verifier sees not only a value but its reliability.

(2) A compliance rules engine. The EU methodology — simple-versus-complex-goods logic, precursor (upstream steel) emissions, and the punitive default-value tables of the 2,400-page implementing regulation⁸ — is encoded as a rules engine that populates the European Commission's communication template and flags each field *actual / default / needs verifier attention*.

(3) A carbon- and price-aware scheduler. Taiwan publishes every generating unit's output at 10-minute resolution as open data²⁶. CarbonPass computes an hourly grid-carbon-intensity curve from that feed, overlays Taipower time-of-use tariffs¹⁹, and optimizes when flexible loads (heat-treatment furnaces, compressors, plating rectifiers) run, subject to order deadlines — a mixed-integer-programming baseline refined by a reinforcement-learning policy.

The architecture is **local-first by design**: parsing and computation run on the factory's own device or premises, and only the outputs the owner chooses to send ever

leave. This directly answers the trade-secret sensitivity that otherwise blocks SME adoption of cloud carbon tools.

2.2 Corresponding SDGs

SDG 8 (Decent Work and Economic Growth) — protecting employment and livelihoods across an export cluster of ~1,800 firms threatened by loss of EU market access. **SDG 9 (Industry, Innovation and Infrastructure)** — equipping traditional manufacturing SMEs with AI-grade data infrastructure. **SDG 12 (Responsible Consumption and Production)** — product-level emissions measurement and verified reduction. **SDG 13 (Climate Action)** — extending measurement and mitigation into the SME layer that national climate targets cannot reach without it.

2.3 Problem to be resolved and its importance

The target population is specific, named, and economically pivotal

Taiwan has 1.716 million SMEs (98.87% of all enterprises), employing 9.19 million people, or 79.3% of the workforce¹¹. The overwhelming majority — shops, services, light logistics — face no carbon obligation. This proposal therefore rejects the inflated “1.7 million SMEs must decarbonize” framing and targets the precisely defined population under hard compulsion today: the ~**2,600 SMEs** that the Ministry of Environment (MOENV) identifies as producers of the CBAM-regulated goods Taiwan exports to the EU¹ (Figure 1).

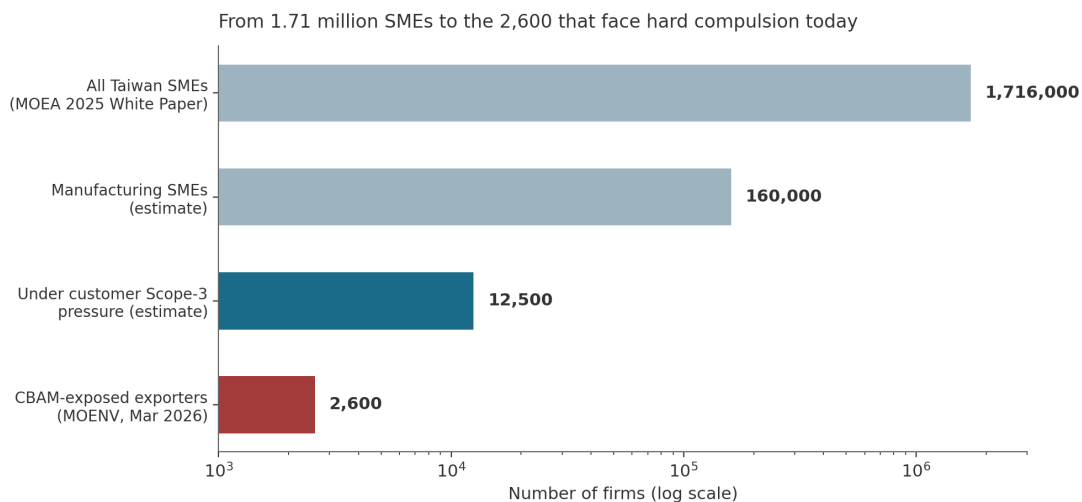


Figure 1: The addressable population is precisely bounded, not inflated: of Taiwan’s 1.716 million SMEs¹¹, the group facing hard compulsion today is the ~2,600 CBAM-exposed exporters named by MOENV¹. The two intermediate bars are this proposal’s estimates of the wider, customer-pressured band.

These firms are concentrated in one place and one product. The Gangshan-Luzhu district of Kaohsiung — the “Kingdom of Screws” — hosts roughly **1,800 fastener manufacturers**, about 700 of them organized under the Taiwan Industrial Fasteners Institute (TIFI)^{12,16}. Their global standing is disproportionate to their size (Figure 2, Table 1): a top-three world exporter holding 10–13% of global production, shipping ~1.2 million tonnes a year, roughly a quarter of it to Europe^{12,13}. Nationally, 51.79% of SMEs are sole proprietorships or family firms and 58.22% have operated more than eight years¹¹ — the demographic profile of the

owners now asked to operate EU-grade data systems.

The "Kingdom of Screws": ~1,800 manufacturers, ~1.2 M tonnes exported (2025), ~1/4 to Europe

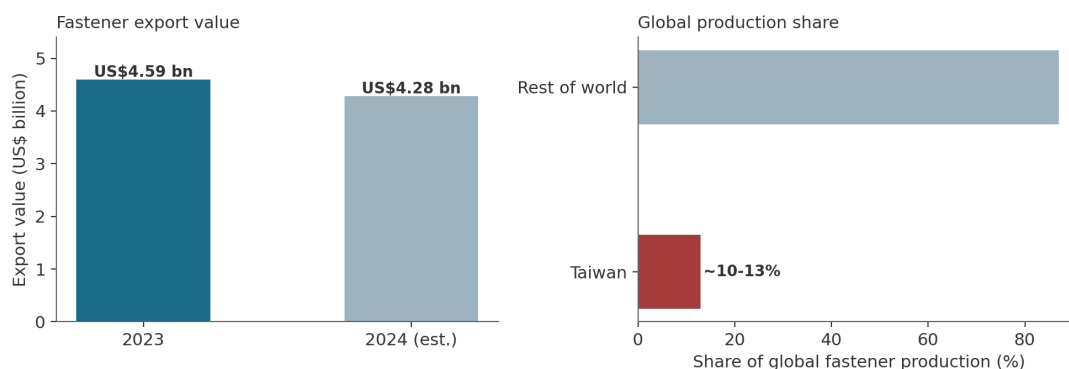


Figure 2: The beachhead sector: export value (left, industry statistics²⁴) and share of world production (right^{12,16}).

Table 1: The Taiwanese fastener sector at a glance

Indicator	Value	Source
Global rank	Top-3 exporter (#2 by value, 2023; #3 by volume, 2025–26)	CW 2023 ¹³ ; KT 2026 ¹²
Share of world production	~10–13%	12,16
Export volume	~1.2 million tonnes (2025)	12
Export value	US\$4.59 bn (2023); ~US\$4.28 bn (2024, est.)	Fastener World ²⁴
Share destined for Europe	~one quarter	13
Manufacturers	~1,800 (mostly SMEs; TIFI ~700 members)	16
Embedded emissions	~2 tCO ₂ e per tonne of product (industry est.)	17

The mechanism: how missing data converts into lost orders

The EU's Carbon Border Adjustment Mechanism (CBAM) entered its definitive, paying phase on 1 January 2026. Importers of covered steel goods — including screws and bolts, customs code CN 7318 — must buy and surrender CBAM certificates priced at the official EU rate: **EUR 75.36 in Q1 and EUR 75.28 in Q2 2026**^{6,15,23}. With verified data, a tonne of fasteners (~2 tCO₂e embedded¹⁷) costs the buyer about **EUR 150**. Without it, the importer must use default values set at the highest observed intensity and marked up 10% in 2026, 20% in 2027, and 30% from 2028; for complex goods, at least 80% of reported emissions must be actual, third-party-verified data^{8,9}. Industry analysis puts the default cost at three to five times the verified cost, reaching **30–50% of product value** on commodity lines¹⁷ (Figure 3).

The 50-tonne de minimis exemption does not protect small suppliers: it is counted cumulatively *per EU importer*, across all of that importer's suppliers combined. MOENV's Climate

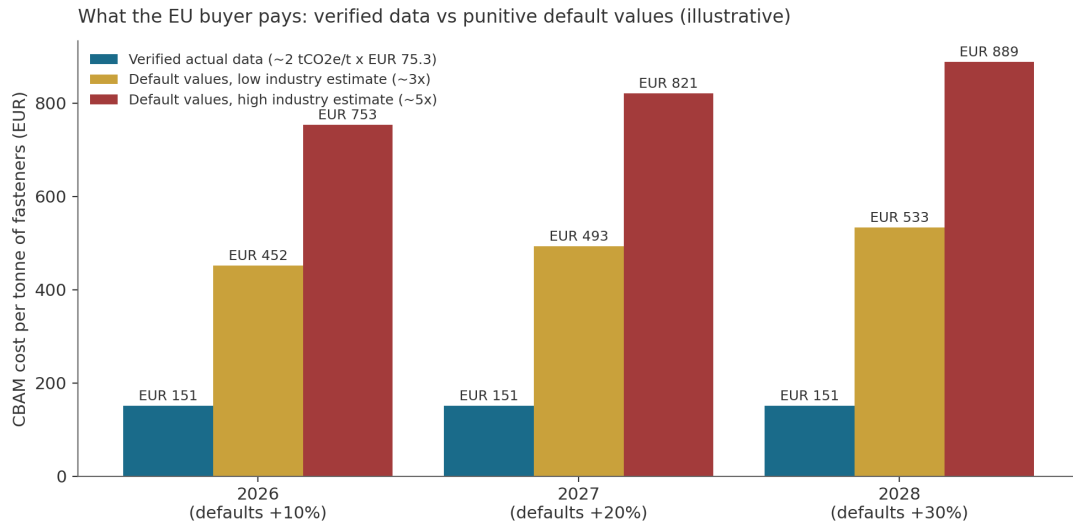


Figure 3: What the EU buyer pays per tonne of Taiwanese fasteners: verified actual data versus punitive defaults, at the official certificate price²³, escalated by the regulation’s automatic markup schedule⁸. The difference is the survival margin of a family factory.

Change Administration has warned publicly that a European distributor buying small lots from several Taiwanese SMEs crosses the threshold and must then demand verified data from all of them¹. The commercial logic is impersonal: an importer facing two otherwise identical suppliers keeps the one whose data pack arrives with the shipment. Sector surveys confirm how exposed the cohort is — **60%** of fast-growing SMEs have completed no carbon inventory, **80%** have no reduction targets, and half cite talent shortage as the binding constraint¹³.

Why the problem compounds, and why it matters nationally

The pressure is a rising floor, not a one-time wave. The default markup escalates automatically to 2028⁸; the European Parliament votes in September 2026 on extending CBAM to ~180 further steel and aluminium product categories from 2028 (Council position adopted June 2026)¹⁰; and the EU’s Digital Product Passport regime begins phasing in machine-readable product-sustainability data from 2027²⁷. Domestically, Taiwan’s own carbon price is live: the first fee cycle closed 31 May 2026, collecting **NT\$4.97 billion from 461 factories run by 240 companies**^{2,3} (Figure 4), with the 25,000-tonne threshold set to fall and an emissions-trading pilot approaching ~2027³¹. These instruments serve legislated targets: net-zero by 2050 and the NDC 3.0 commitment of a **38%±2% cut by 2035**²¹ (Figure 5) — targets that cannot be met by the ~500 directly regulated large emitters alone, and that require measurement capillaries reaching into the SME layer this proposal serves.

2.4 Related stakeholders and their roles

The **target audience is the fastener SME owner** — chosen because this group carries live legal-commercial compulsion, has the lowest digital readiness, and is geographically and industrially concentrated enough for a focused pilot to generalize. The remaining stakeholders form the adoption and verification chain around that user.

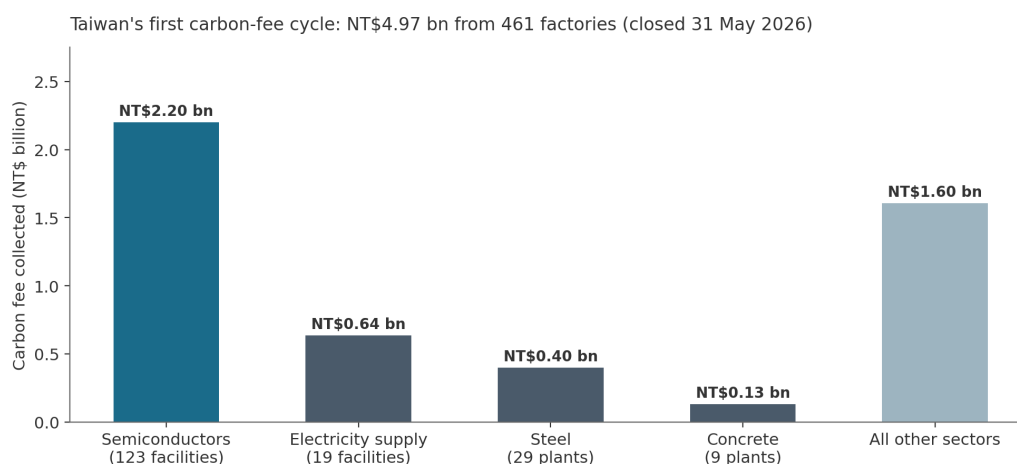


Figure 4: Taiwan's first carbon-fee cycle, closed 31 May 2026 (MOENV via Focus Taiwan / Taipei Times^{2,3}): NT\$4.97 billion, with semiconductors paying 44%. The domestic carbon price is live and set to broaden.

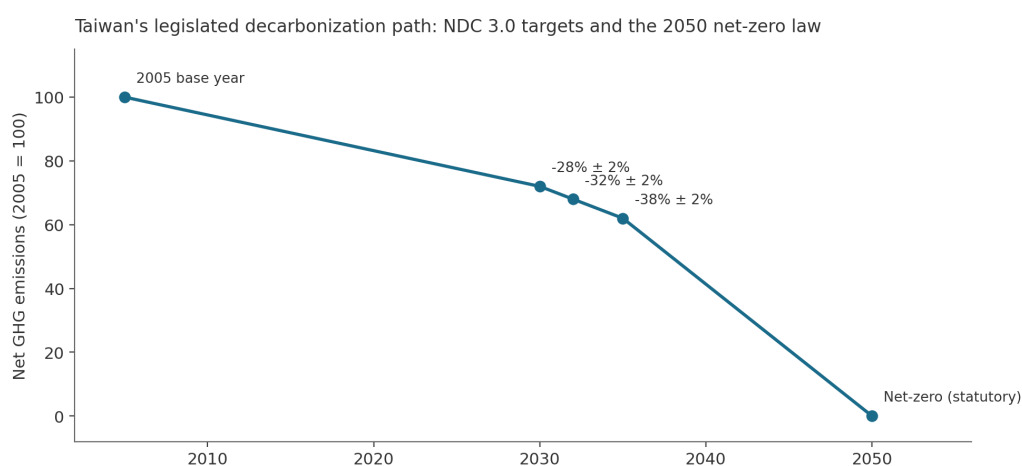


Figure 5: Taiwan's legislated emissions path: the NDC 3.0 targets (MOENV, 2025²¹) and the statutory 2050 net-zero goal. Reaching them requires extending measurement into the SME base.

Table 2: Stakeholders, roles, and benefits (target audience highlighted)

Stakeholder	Role and benefit
Fastener SME owners (target audience)	Primary users: retain EU customers, cut power costs, and gain a standing carbon capability in their own language
EU importers and their declarants	Receive template-ready supplier data; avoid punitive default costs ⁸
TIFI (~700 members) and MIRDC	Pilot recruitment, domain validation, and the channel for cluster-scale rollout ¹⁶
MOENV / Climate Change Administration	Their CBAM help platform ¹ gains the computation layer it currently lacks; SME carbon-fee readiness rises
MOEA / SMESA	The CAAS education stack ¹⁸ gains an execution tool for SME competitiveness
Taiwan Power Company	Its open data and AMI infrastructure ^{19,26} become directly value-generating for small industry
Accredited verifiers	Receive structured, uncertainty-annotated files rather than raw paper — faster, cheaper verification
Banks (green finance)	Verified SME carbon data for sustainability-linked lending

2.5 Description of solution functions

How the target audience uses it

Mr. Lin, a 61-year-old owner in Gangshan, opens LINE. The assistant greets him in Taiwanese and asks for a photo of last month's Taipower bill. He photographs it, then the steel wire-rod invoices, and answers three plain questions about production. Within about ninety seconds he receives: a **carbon data pack** for his screws, per customs code, in the European Commission's template and flagged line-by-line for the verifier; a **one-screen argument** he can forward to his buyer — *your cost with my data: $\approx$ EUR 150/t; without it: EUR 450-750/t and rising*; and a **scheduling suggestion** — shift the heat-treatment furnace to the 02:00-06:00 window this month, for an estimated saving in NT\$ on the bill and in kilograms of CO₂, both of which flow into next quarter's pack automatically. No web portal, no consultant, and no data leaving his building.

How the problem is solved, stage by stage

Table 3: The CarbonPass pipeline: from shoebox documents to market access

Stage	What happens
Inputs the SME already has	Taipower bills, material invoices, production records — photographed in LINE, in Mandarin/Taiwanese
Document intelligence + allocation	Vision-language parsing of messy, mixed-format documents; facility energy allocated across product lines and CN codes with quantified uncertainty
Compliance rules engine	EU methodology (simple/complex goods, precursors, default tables ⁸) encoded into field-level guidance and the Commission's template
Grid-aware scheduler	Hourly grid carbon intensity from Taipower's open 10-minute feed ²⁶ + time-of-use tariffs ¹⁹ → MILP + reinforcement-learning shift plans
Outputs	CBAM data pack per CN code; organizational inventory (by-product); buyer-cost delta screen; monitored before/after reduction ledger

The two modules reinforce each other. The Data-Pack Engine (Module 1) answers the immediate compliance demand; the Grid-Aware Scheduler (Module 2) then lowers real consumption, and every avoided tonne shrinks the next data pack. The reduction ledger is structured to the baseline-and-monitoring logic of ISO 14064-2 — structured to the standard’s logic, with certification remaining the accredited verifier’s act. The conversational LINE interface (Module 3) is not cosmetic: it is the mechanism by which an analog-native owner participates at all, and is therefore the operative form of digital inclusion in this project.

2.6 Differences from existing solutions

Taiwan’s current ecosystem is an *awareness* stack; CarbonPass is the missing *execution* layer. The audit below was verified instrument by instrument in July 2026.

Table 4: Capability audit: what exists, and what none of it does

Instrument	What it does	What it does not do
CAAS “SME Carbon Reduction Service Station” (MOEA) ¹⁸	Courses, articles, consultant referrals	No computation; not a reporting tool
IDB Carbon Emissions Calculator (MOEA) ²⁰	Rough annual company-level CO ₂ e from bills	No product-level allocation, no CN codes, no CBAM format
MOENV CBAM Help Platform (Mar 2026) ¹	Information and consultation	Explicitly informational; computes nothing
Taipower app / eBill ¹⁹	Consumption charts, tariff simulation	No emissions conversion; no compliance artifacts
tre100 pledge platform (CIER) ²⁵	Renewable-energy commitments registry	No measurement tooling
Consultants / ESG software ³⁰	Real one-off ISO certificates	Per-engagement pricing; cannot serve recurring per-product demand across 2,600 firms

The difference in one sentence: CarbonPass is the only tool that converts SME-native documents into importer-ready, verifier-structured, per-CN-code emissions data — repeatedly, in the owner’s own language, at near-zero marginal cost — and it is designed to *integrate* with the government’s existing platforms rather than compete with them.

2.7 Expected benefits (quantified)

3 Maturity of the Project

3.1 Development stage

CarbonPass is an early prototype purpose-built for this competition; it has not been commercialized, and the modification-rule provisions therefore do not apply. At submission the following is demonstrable:

- **Module 1 (Data-Pack Engine)** runs end-to-end on a three-firm mock corpus built from real Taiwanese Taipower bill and Ministry of Finance e-invoice formats, producing a per-CN-code data pack in the European Commission template with per-line uncertainty.
- **Module 2 (Grid-Aware Scheduler)** runs on a synthetic fastener-plant load profile

Table 5: Expected benefits, with basis of estimate

Benefit	Quantification	Basis
CBAM cost avoided per tonne	≈EUR 300-600/t (verified ≈150 vs. defaults 450-750, rising with markups)	Official price ²³ ; industry multiples ¹⁷
Compliance labor per data pack	Hours instead of consultant-weeks; recurring at near-zero marginal cost	Cluster ISO precedents priced per engagement ³⁰
Energy cost reduction via load shifting	Peak/off-peak tariff spread of several× on shiftable loads (est. 5-15% of bill for furnace-heavy plants; measured in pilot)	Taipower TOU schedules ¹⁹
Emissions reduction	Measured on-meter, logged to ledger; each avoided tonne also cuts the buyer's certificate cost	Grid EF 0.474 kgCO ₂ e/kWh ²²
Population served at beachhead	2,600 CBAM-exposed SMEs within a ~1,800-firm, top-3 world export cluster	MOENV ^{1, 12, 16}
National alignment	Extends measurement into the SME layer, supporting NDC 3.0 (−38%±2% by 2035)	²¹

against the *live* open 10-minute grid feed, computing hourly carbon intensity and an optimized shift plan²⁶.

- **Module 3 (Inclusion Interface)** is an operational LINE conversational front end in Mandarin/Taiwanese.

The proof of concept requires no hardware, no paid API, and no partnership to run — it is built entirely on open data (Section 5), which makes its feasibility independently checkable by the jury.

3.2 Project adjustments

Not applicable: the project is original to this competition and has not been sold or awarded elsewhere. No 50% modification was required.

3.3 Demonstration

A demonstration video accompanies the online application, showing the full path from photographed documents to a generated data pack, the buyer-cost delta screen, and a grid-aware schedule computed from the live open grid feed. *[Video link entered in application form field 3.3.]*

3.4 Existing validation fields

Validation is planned within the competition's own mentorship machinery and involves the public sector. During the mentorship period, outreach is directed to a TIFI member factory in the Gangshan cluster for a real pilot, with the Metal Industries Research & Development Centre (MIRDC) and Kaohsiung Environmental Protection Bureau's voluntary-reduction advisory team as domain and public-sector validation partners. *These are validation targets being approached, not yet secured commitments*, and are stated as such.

3.5 Existing validation results and verification plan

At submission the quantified results are those produced on the mock corpus and synthetic-plus-live-feed pipeline (Section 3.1); the field-measured results are generated during mentorship and presented at the final review. The plan maps directly onto the official calendar²⁹:

Table 6: Implementation and verification milestones mapped to the official schedule

Window (Handbook)	Milestone
By 31 Jul 2026 (submission)	Proof of concept v0 + demonstration video + this proposal; validation outreach opened to TIFI, MIRDC, and Kaohsiung EPB
6–16 Aug (preliminary review)	Proposal and working prototype under evaluation
Mid-Sep – mid-Oct (mentorship)	One pilot factory in Gangshan: real bills in, real data pack out; pack structure reviewed with an MIRDC engineer or accredited verifier; one flexible load shifted and measured on-meter; weekly progress log
Late Oct (final review)	Live demonstration: data pack generated on stage from raw documents; live grid-aware schedule; pilot before/after ledger; market outlook updated with the September European Parliament vote ¹⁰

BOUNDARIES STATED UP FRONT

CarbonPass *prepares* verification; it does not replace the accredited verifier — at least 80% of reported emissions must be third-party verified⁸. Preferential carbon-fee rates belong to large regulated emitters, not SMEs; the SME's incentives are customer retention, cost pass-through, energy savings, and readiness⁴. The 2028 scope extension is pending legislation and is treated as a tracked roadmap event, not a booked benefit¹⁰.

4 Future Plans

4.1 Promotion and adoption plan

The path to reach the target population runs through the institutions the cluster already trusts rather than through direct marketing to individual factories. TIFI's ~700 member firms and MIRDC's subsidized counseling channel are the primary distribution routes; a single verified Gangshan pilot, documented, becomes the reference case that the association can propagate. In parallel, the project seeks integration of its computation layer into MOENV's existing CBAM help platform and SMESA's CAAS service network^{1,18}, so that CarbonPass reaches SMEs as an extension of services they already consult, at the moment of need. Public visibility from the competition itself, and the openly published datasets (Section 5), further lower the cost of discovery.

4.2 Engaged collaboration units

At submission, collaboration is at the outreach stage and is stated honestly as such: TIFI member firms, MIRDC, and Kaohsiung EPB's voluntary-reduction advisory team are being approached as pilot and validation partners for the mentorship period. The competition's own public-private matchmaking and field-validation support are expected to formalize these relationships for advancing teams²⁹.

4.3 Future planned collaboration units

Table 7: Planned collaboration and the expansion sequence

Horizon	Partners	What happens
2026–2027 (Taiwan beachhead)	TIFI, MIRDC, MOENV, SMESA/MOEA	Harden the pilot into cluster-scale roll-out; propose integration into the MOENV CBAM platform and CAAS network; publish open datasets and open-source components ^{1,18}
2027–2028 (regulatory expansion)	EU importers; standards bodies	If the European Parliament confirms the downstream extension, onboard newly covered categories — springs, wire, hand tools, household metal articles — from the same engine ¹⁰ ; extend outputs toward Digital Product Passport payloads ²⁷
2027 onward (domestic breadth)	MOENV; FSC (green finance); banks	Serve SMEs entering Taiwan’s lowered carbon-fee threshold and ETS pilot era ³¹ ; provide bank-ready carbon data for sustainability-linked lending
2028+ (international)	SME-exporting economies in Asia	Port the architecture to Vietnam, Thailand, Malaysia, and India, which face the same mechanism with thinner tooling

The expansion is one engine with a widening catalogue: fasteners (2026, live compulsion) → downstream steel and aluminium categories (2028, pending) → Digital Product Passport sectors (2027–2030) → domestic carbon-fee entrants (2027 onward). The same open-data grid signal, local-first document AI, and EU-format output serve every stage.

Beyond Taiwan. The carbon-data wall is not Taiwan’s alone. Every SME-exporting economy faces the same EU mechanism, most with weaker digital infrastructure than Taiwan’s. CarbonPass is portable by design — swap the national grid feed, emission coefficients, and tariff tables, and the same architecture serves another country’s exporters. A Taiwanese answer to a global problem: digital inclusion delivered as trade capability.

5 Open Data

Open data is the founding ethos of the Presidential Hackathon, and the proof of concept is built on it end-to-end, requiring no proprietary data source.

5.1 Open-data sources used

5.2 Suggestions for modifying open data

Working with these datasets surfaced three concrete improvements that would benefit any exporter, not only this project: (1) **publish an official hourly grid-carbon-intensity series** — today it must be derived from the 10-minute generation-by-unit feed; (2) **publish MOENV emission coefficients mapped to customs (CN/HS) codes**, so any exporter can locate the factors relevant to a specific tariff line; and (3) **provide a machine-readable Taipower bill format**, enabling consented, automated ingestion without manual transcrip-

Table 8: Open data used in the proof of concept (links are clickable)

Resource	Access	Role
MOENV carbon-footprint emission coefficients	data.gov.tw #28176 (OpenAPI)	Material / fuel / process emission factors
Taiwan product carbon-footprint registry	data.gov.tw #8992	Benchmarks and sanity checks
Electricity emission factor, 0.474 kgCO ₂ e/kWh (2024)	MOEA Energy Administration	Scope-2 and scheduler math
Taipower generation by unit, 10-minute	#8931 (real-time) , #37331 (historical)	Hourly grid-carbon-intensity curve
Taipower time-of-use rate schedules	taipower.com.tw	Price signal for the scheduler
CBAM methodology, default values (Reg. (EU) 2025/2621; DG TAXUD templates)	EUR-Lex / DG TAXUD	Output schema; CN 7318 default rows; cost-delta screen
Official CBAM certificate prices (quarterly)	European Commission	Live cost math (EUR 75.36 Q1 / 75.28 Q2 2026)
EU-Taiwan trade flows for CN 7318	Eurostat Comext; Taiwan MOF customs	Impact quantification

tion.

5.3 Provision of open data (given back)

CarbonPass publishes back, under an open license: (1) a **cleaned, documented hourly Taiwan grid-carbon-intensity dataset** derived from [#8931](#) / [#37331](#), useful to any researcher or firm doing carbon-aware scheduling; and (2) an **anonymized fastener-sector embedded-emissions benchmark** from the pilot, useful to the whole cluster and to verifiers. Both are organized with documentation and stable links for others to reuse.

6 Open-Source Software Design

The project intends to release its highest public-value components — the CBAM compliance rules engine and the grid-intensity pipeline — under an open license, and to build on open-source foundations (open vision-language models, open optimization libraries) for the proof of concept. The team is willing to allow future open-source use of these components, consistent with the competition’s encouragement of open-source contribution. (*Per the Handbook, this section does not affect the project score*²⁹.)

When the AI era builds a wall of data around the world’s markets, digital inclusion means handing every small factory a door. CarbonPass is that door — a photograph of an electricity bill, met by the right AI, keeping two thousand six hundred family factories, and the district they built, inside the world economy.

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